Knowledge index: topic

Generated by `python scripts/kb.py generate-indices`. Tags and architecture labels describe discussion scope; they do not establish measured transfer.

architecture

- Compute, parameters and a constrained hero budget — `foundation-compute-accounting`; inferred.
- Width, depth, μP, and training hyperparameters: what actual ladders do — `guide-architecture-evidence`; catalogued companion guide.
- MoE architecture choices: Qwen3, OLMoE and Marin — `guide-moe-families`; catalogued companion guide.
- SMELT v3: what two-pass MoE looping actually establishes — `guide-smelt-looped-moe`; catalogued companion guide.

batch-size

- Width, batch and horizon: a conditional hyperparameter-transfer protocol — `technique-hyperparameter-transfer`; inferred.

compute

- Compute, parameters and a constrained hero budget — `foundation-compute-accounting`; inferred.
- Candidate v1: a smaller overtrained looped MoE on 128 H100s — `guide-hero-h100-candidate`; catalogued companion guide.
- Data-first candidate plan: 96 H100s for 30 days — `guide-plan-96-h100`; catalogued companion guide.
- Offline workbench: public records and analytical fixtures — `recipe-offline-workbench`; inferred.

cyber

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.

data

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Adding high-quality cyber data to a Marin-style corpus — `guide-cyber-data`; catalogued companion guide.
- Marin replication: PRs, code, data and result receipts — `guide-marin-artifact-lineage`; catalogued companion guide.
- Replicating Marin Datakit: a code-level data-processing manual — `guide-marin-code-walkthrough`; catalogued companion guide.
- Building our corpus with Marin's data-engineering approach — `guide-marin-data-engineering`; catalogued companion guide.
- Data-mixture experiments: supply, exposure, controls and transfer — `technique-data-mixture-design`; inferred.

evaluation

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Checkpoint evaluation: from ladder loss to usable-model evidence — `guide-checkpoint-evaluation`; catalogued companion guide.

- Marin evaluation: pretraining, mid-training and post-training — `guide-marin-evaluation`; catalogued companion guide.
- Diagnosing a failed scaling or training decision — `pattern-failure-diagnosis`; inferred.
- Evaluation contracts: losses, capabilities, contexts and failures — `technique-evaluation-contract`; inferred.

governance

- Pretraining knowledge map and reading dependencies — `foundation-topic-map`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Knowledge coverage, unresolved inputs and review dates — `governance-coverage`; inferred.
- A rigorous pretraining knowledge base: evidence and reproduction — `governance-evidence-standard`; inferred.
- Understanding scientific pretraining ladders — `guide-index`; catalogued companion guide.

holdouts

- Identifiable scaling fits and independent larger holdouts — `technique-scaling-fit`; inferred.

identifiability

- Identifiable scaling fits and independent larger holdouts — `technique-scaling-fit`; inferred.

kernelwiki

- A rigorous pretraining knowledge base: evidence and reproduction — `governance-evidence-standard`; inferred.

learning-rate

- Width, batch and horizon: a conditional hyperparameter-transfer protocol — `technique-hyperparameter-transfer`; inferred.

literature

- Pretraining knowledge map and reading dependencies — `foundation-topic-map`; inferred.
- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Marin reading map — `guide-MARIN-READING`; catalogued companion guide.
- Reference collection status — `guide-README`; catalogued companion guide.
- Source catalogue — `guide-SOURCES`; catalogued companion guide.
- The science of pretraining: annotated deep-dive notes — `guide-deep-dive-notes`; catalogued companion guide.
- StepFun Part II: Farseer's loss surface and allocation experiments — `guide-farseer`; catalogued companion guide.
- Reading Jiaxuan's scientific scaling ladder blog — `guide-jiaxuan-reading-guide`; catalogued companion guide.
- Marin in depth: corpus pipelines, mixture proxies, and Delphi — `guide-marin-pretraining`; catalogued companion guide.
- Nemotron pretraining: Nano, Super, Ultra, and Nemotron-CC — `guide-nemotron-pretraining`; catalogued companion guide.
- Review of the deep-dive notes against primary evidence — `guide-notes-review`; catalogued companion guide.
- OLMo in depth: from corpus construction to validated training — `guide-olmo-pretraining`; catalogued companion guide.
- Power Lines: batch size, weight decay and long-horizon training — `guide-power-lines`; catalogued companion guide.

- How to Build a Scientific Scaling Ladder — `guide-scientific-scaling-ladder`; catalogued companion guide.
- Source housekeeping: what we have and where to find it — `guide-source-guide`; catalogued companion guide.
- StepFun Part I: Step Law, experiment by experiment — `guide-step-law`; catalogued companion guide.

long-context

- Evaluation contracts: losses, capabilities, contexts and failures — `technique-evaluation-contract`; inferred.

marin

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Offline workbench: public records and analytical fixtures — `recipe-offline-workbench`; inferred.
- Data-mixture experiments: supply, exposure, controls and transfer — `technique-data-mixture-design`; inferred.

midtraining

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.

mixture

- Data-mixture experiments: supply, exposure, controls and transfer — `technique-data-mixture-design`; inferred.

moe

- Compute, parameters and a constrained hero budget — `foundation-compute-accounting`; inferred.

mup

- Width, batch and horizon: a conditional hyperparameter-transfer protocol — `technique-hyperparameter-transfer`; inferred.

optimization

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.
- Replicating Marin's optimizer tuning and transfer — `guide-marin-hyperparameter-search`; catalogued companion guide.
- Diagnosing a failed scaling or training decision — `pattern-failure-diagnosis`; inferred.
- Width, batch and horizon: a conditional hyperparameter-transfer protocol — `technique-hyperparameter-transfer`; inferred.

posttraining

- Marin replication: a stage-by-stage evidence and acceptance contract — `case-marin-reproduction`; inferred.

provenance

- Scoped scientific claim ledger — `governance-claim-ledger`; inferred.
- Knowledge coverage, unresolved inputs and review dates — `governance-coverage`; inferred.
- A rigorous pretraining knowledge base: evidence and reproduction — `governance-evidence-standard`; inferred.

recurrence

- Compute, parameters and a constrained hero budget — `foundation-compute-accounting`; inferred.

repetition

- Data-mixture experiments: supply, exposure, controls and transfer — `technique-data-mixture-design`; inferred.

reproducibility

- Knowledge coverage, unresolved inputs and review dates — `governance-coverage`; inferred.
- A rigorous pretraining knowledge base: evidence and reproduction — `governance-evidence-standard`; inferred.
- Diagnosing a failed scaling or training decision — `pattern-failure-diagnosis`; inferred.
- Offline workbench: public records and analytical fixtures — `recipe-offline-workbench`; inferred.

scaling

- Pretraining knowledge map and reading dependencies — `foundation-topic-map`; inferred.
- Scope: what a scaling ladder predicts — `guide-01-scope`; catalogued companion guide.
- Objectives, evaluation, and acceptance criteria — `guide-02-objectives`; catalogued companion guide.
- Measurement: parameters, FLOPs, tokens, and loss — `guide-03-measurement`; catalogued companion guide.
- Architecture: a controlled baseline and structural ladders — `guide-04-architecture`; catalogued companion guide.
- Experimental grid, checkpoints, and compute budget — `guide-05-grid-budget`; catalogued companion guide.
- Optimizers, hyperparameter transfer, and annealed comparisons — `guide-06-optimizers`; catalogued companion guide.
- Data experiments: source value, mixture weights, and repetition — `guide-07-data-mixture`; catalogued companion guide.
- Fitting loss surfaces and diagnosing extrapolation — `guide-08-fitting`; catalogued companion guide.
- Predicting downstream capabilities — `guide-09-downstream`; catalogued companion guide.
- Holdouts, stability, and production decisions — `guide-10-validation`; catalogued companion guide.
- Staged training and long-context ladders — `guide-11-specialized`; catalogued companion guide.
- Execution records, deviations, and ladder maintenance — `guide-12-execution`; catalogued companion guide.
- Public examples: Chinchilla, optimizer grids, and Delphi — `guide-13-public-ladders`; catalogued companion guide.
- From predicted loss to a validated capability forecast — `guide-capabilities-and-validation`; catalogued companion guide.
- Turn mixture weights into auditable data exposure — `guide-data-and-repetition`; catalogued companion guide.
- Starting with Nemotron-CC or Marin data: the complete pretraining design — `guide-from-data-to-pretraining`; catalogued companion guide.
- Design a ladder that identifies the question you ask — `guide-grid-and-inference`; catalogued companion guide.
- Industry workflow: auditing the Jiaxuan conversation — `guide-industry-practice-audit`; catalogued companion guide.
- From a loss law to a model-size decision — `guide-loss-and-allocation`; catalogued companion guide.

- Count parameters, tokens, and executed training work — `guide-measurement-and-compute`; catalogued companion guide.
- Learning-rate schedules, batch size, and optimizer memory — `guide-schedules-and-timescales`; catalogued companion guide.
- The complete workflow for scaling-law experiments — `guide-workflow`; catalogued companion guide.
- Identifiable scaling fits and independent larger holdouts — `technique-scaling-fit`; inferred.

step-law

- Offline workbench: public records and analytical fixtures — `recipe-offline-workbench`; inferred.

systems

- Diagnosing a failed scaling or training decision — `pattern-failure-diagnosis`; inferred.

tokenizer

- Evaluation contracts: losses, capabilities, contexts and failures — `technique-evaluation-contract`; inferred.

training

- Marin mid-training and post-training: concrete experiments — `guide-marin-midtraining-posttraining`; catalogued companion guide.
- Training and tuning with TorchTitan and Megatron-LM — `guide-trainer-playbook`; catalogued companion guide.

uncertainty

- Evaluation contracts: losses, capabilities, contexts and failures — `technique-evaluation-contract`; inferred.
- Identifiable scaling fits and independent larger holdouts — `technique-scaling-fit`; inferred.

weight-decay

- Width, batch and horizon: a conditional hyperparameter-transfer protocol — `technique-hyperparameter-transfer`; inferred.